

AI-Powered Business Intelligence & Sales Analytics Dashboard

Data Analytics / Business Intelligence Project

Abstract

This project develops an AI-Powered Business Intelligence and Sales Analytics Dashboard using MySQL, Tableau and ChatGPT. Retail sales data was analyzed to understand revenue, profitability, customer behavior, regional trends, product performance and discount impact. SQL queries were used to calculate KPIs while Tableau dashboards provided interactive visualizations. The solution enables managers to monitor performance and make data-driven decisions. This project develops an AI-Powered Business Intelligence and Sales Analytics Dashboard using MySQL, Tableau and ChatGPT. Retail sales data was analyzed to understand revenue, profitability, customer behavior, regional trends, product performance and discount impact. SQL queries were used to calculate KPIs while Tableau dashboards provided interactive visualizations. The solution enables managers to monitor performance and make data-driven decisions. This project develops an AI-Powered Business Intelligence and Sales Analytics Dashboard using MySQL, Tableau and ChatGPT. Retail sales data was analyzed to understand revenue, profitability, customer behavior, regional trends, product performance and discount impact. SQL queries were used to calculate KPIs while Tableau dashboards provided interactive visualizations. The solution enables managers to monitor performance and make data-driven decisions.

Introduction

Business Intelligence helps organizations transform raw data into meaningful insights. Modern organizations generate large volumes of transactional data that require analysis. Dashboard reporting improves decision-making through KPI monitoring and trend analysis. Business Intelligence helps organizations transform raw data into meaningful insights. Modern organizations generate large volumes of transactional data that require analysis. Dashboard reporting improves decision-making through KPI monitoring and trend analysis. Business Intelligence helps organizations transform raw data into meaningful insights. Modern organizations generate large volumes of transactional data that require analysis. Dashboard reporting improves decision-making through KPI monitoring and trend analysis.

Problem Statement

Businesses struggle to extract insights from raw sales data. Manual reporting is time consuming and makes it difficult to identify profitable regions, products and

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Existing System

Traditional spreadsheet-based reporting lacks interactivity, requires manual effort and delays decision making. Traditional spreadsheet-based reporting lacks interactivity, requires manual effort and delays decision making. Traditional spreadsheet-based reporting lacks interactivity, requires manual effort and delays decision making.

Proposed System

The proposed system uses MySQL for analysis, Tableau for visualization and ChatGPT for insight generation and documentation support. The proposed system uses MySQL for analysis, Tableau for visualization and ChatGPT for insight generation and documentation support. The proposed system uses MySQL for analysis, Tableau for visualization and ChatGPT for insight generation and documentation support.

Objectives

Analyze sales, profit, customers, products, regions and discounts. Build dashboards and generate business recommendations. Analyze sales, profit, customers, products, regions and discounts. Build dashboards and generate business recommendations. Analyze sales, profit, customers, products, regions and discounts. Build dashboards and generate business recommendations.

Dataset Description

The Sample Superstore dataset contains approximately 9,994 records and includes Order ID, Customer Name, Region, State, Category, Product Name, Sales, Profit, Quantity and Discount. The Sample Superstore dataset contains approximately 9,994 records and includes Order ID, Customer Name, Region, State, Category, Product Name, Sales, Profit, Quantity and Discount. The Sample Superstore dataset contains approximately 9,994 records and includes Order ID, Customer Name, Region, State, Category, Product Name, Sales, Profit, Quantity and Discount.

Methodology

Business Understanding → Data Preparation → SQL Analysis → KPI Identification → Dashboard Development → Insight Generation → Recommendations. Business Understanding → Data Preparation → SQL Analysis → KPI Identification → Dashboard

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Results

The AI-Powered Business Intelligence & Sales Analytics Dashboard successfully analysed retail sales data and generated valuable business insights through SQL and Tableau.

Key KPI Results

- **Total Sales:** \$2.29 Million
- **Total Profit:** \$282.86 Thousand
- **Total Orders:** 4,931
- **Profit Margin:** 12.45%
- **Total Customers:** 793

Executive Overview Findings

- Business maintained strong sales and profitability.
- Sales and profit increased significantly during the final quarter.
- November and December recorded the strongest performance.

Regional Performance Findings

- **West Region** generated the highest sales and profit.
- **East Region** ranked second in performance.
- **California** was the top-performing state in both sales and profit.
- South region showed growth opportunities.

Product Performance Findings

- **Technology** was the highest revenue and profit-generating category.
- **Canon imageCLASS 2200 Advanced Copier** was the top-selling product.
- **Staples** was the most frequently purchased product.
- Some products generated losses despite strong sales volume.

Customer & Segment Findings

- **Consumer Segment** contributed the highest sales and profit.
- A small group of customers generated a significant share of revenue.
- Customer retention can further improve business growth.

Discount & Profitability Findings

- Higher discounts reduced profitability.
- No-discount and low-discount transactions generated the highest profits.
- Excessive discounting negatively affected profit margins.

Business Insights

The dashboard analysis provided several important insights regarding the company's sales performance, profitability, customer behaviour, product trends, and regional performance. These insights help management understand the key drivers of business success and identify areas that require improvement.

- **Regional Performance Insight**

The **West Region** emerged as the strongest-performing region, generating the highest sales and profit among all regions. This indicates strong customer demand, effective sales strategies, and a profitable product mix within the region. In contrast, the **South Region** recorded the lowest sales and profitability, highlighting potential opportunities for market expansion and targeted promotional activities. The company can improve overall performance by adopting successful practices from the West Region and implementing region-specific marketing strategies in underperforming areas.

- **State Performance Insight**

Among all states, **California** contributed the highest sales and profit, making it the most valuable market for the business. The state's strong performance reflects a large customer base and consistent purchasing activity. Other states such as **New York, Texas, and Washington** also contributed significantly to revenue generation. These findings suggest that the organization should continue investing in high-performing states while exploring growth opportunities in emerging markets to reduce dependency on a limited number of locations.

- **Product Performance Insight**

The analysis revealed that the **Technology Category** generated the highest revenue and profit compared to Furniture and Office Supplies. Technology products generally offer higher selling prices and better profit margins, making them major contributors to business growth. The findings indicate that expanding technology product offerings and increasing promotional efforts for high-performing products can further enhance revenue and profitability.

- **Top Product Insight**

The **Canon imageCLASS 2200 Advanced Copier** was identified as the highest revenue-generating product in the dataset. This demonstrates that premium products can contribute significantly to overall sales performance. The company should ensure adequate inventory levels, improve product visibility, and focus marketing efforts on top-performing products to maximize revenue opportunities.

- **Customer Segment Insight**

The **Consumer Segment** contributed the largest share of total sales and profit, making it the most important customer segment for the business. The segment's strong performance indicates high customer engagement and purchasing frequency. Focusing on customer satisfaction, loyalty programs, and personalized marketing initiatives can help strengthen relationships with this valuable customer group and increase long-term profitability.

- **Customer Behaviour Insight**

The analysis showed that a relatively small group of customers generated a substantial portion of total revenue and profit. These high-value customers play a critical role in business success and should be considered strategic assets. Retaining these customers through exclusive offers, personalized services, and customer relationship management strategies can significantly improve customer lifetime value and business stability.

- **Discount and Profitability Insight**

One of the most important findings from the dashboard was the relationship between **discounts and profitability**. The analysis clearly showed that higher discount levels resulted in lower profit margins and, in some cases, financial losses. While discounts can help increase sales volume, excessive discounting negatively impacts overall profitability. Therefore, the company should adopt a more controlled discount strategy that balances customer acquisition with profit generation.

- **Profitability Insight**

The company generated more than **\$2.29 million in sales** and over **\$282 thousand in profit**, resulting in a healthy **profit margin of 12.45%**. This indicates that the business is operating profitably and maintaining a sustainable financial position. However, opportunities still exist to improve profitability by optimizing pricing strategies, reducing unnecessary discounts, and focusing on high-margin products and customers.

Overall Business Insight

Overall, the dashboard successfully identified the key factors driving business performance. The **West Region, California, Technology Category**, and **Consumer Segment** were found to be the primary contributors to sales and profit. At the same time, the analysis highlighted the negative impact of excessive discounting on profitability. By focusing on high-

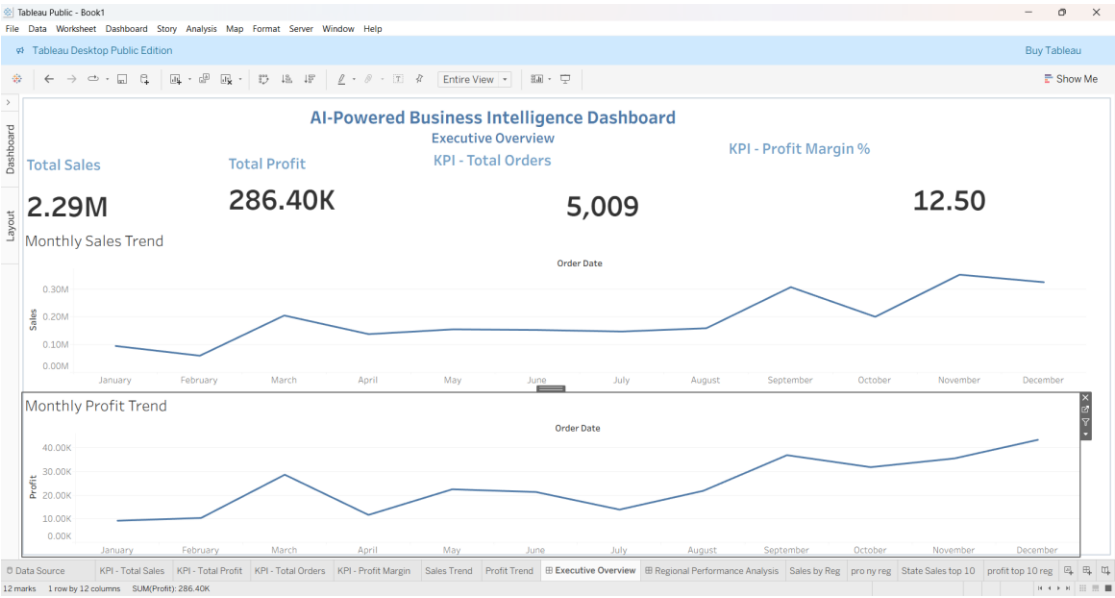
performing regions, products, and customer segments while optimizing discount strategies, the organization can improve decision-making, increase profitability, and support sustainable long-term growth.

Conclusion

The project successfully demonstrates Business Intelligence concepts using SQL and Tableau. It converts raw data into actionable insights for management. The project successfully demonstrates Business Intelligence concepts using SQL and Tableau. It converts raw data into actionable insights for management. The project successfully demonstrates Business Intelligence concepts using SQL and Tableau. It converts raw data into actionable insights for management.

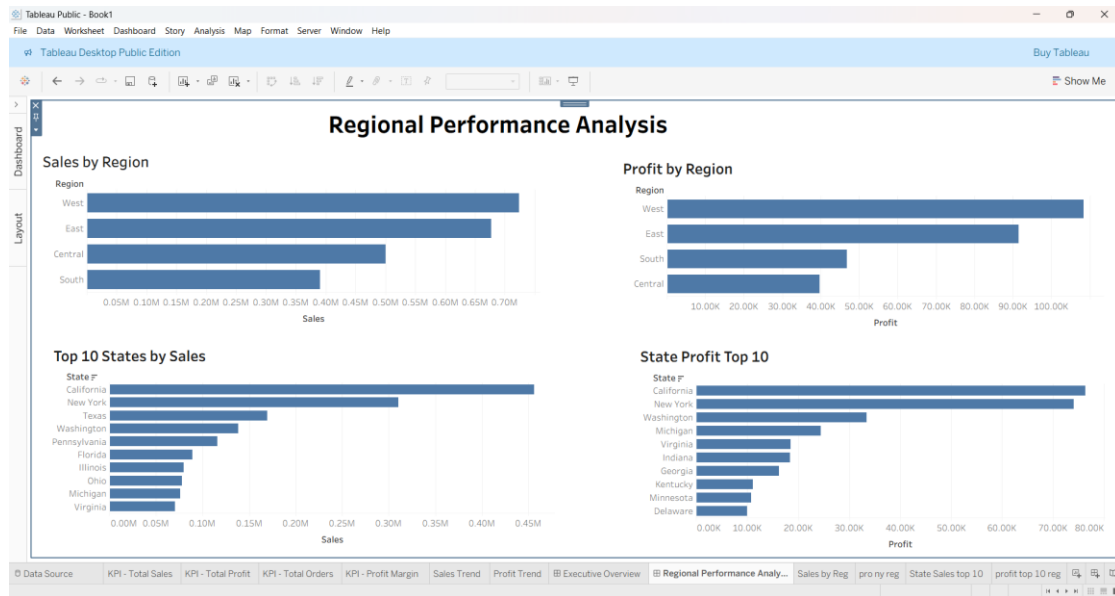
Dashboard Screenshots and Analysis

Executive Overview Dashboard



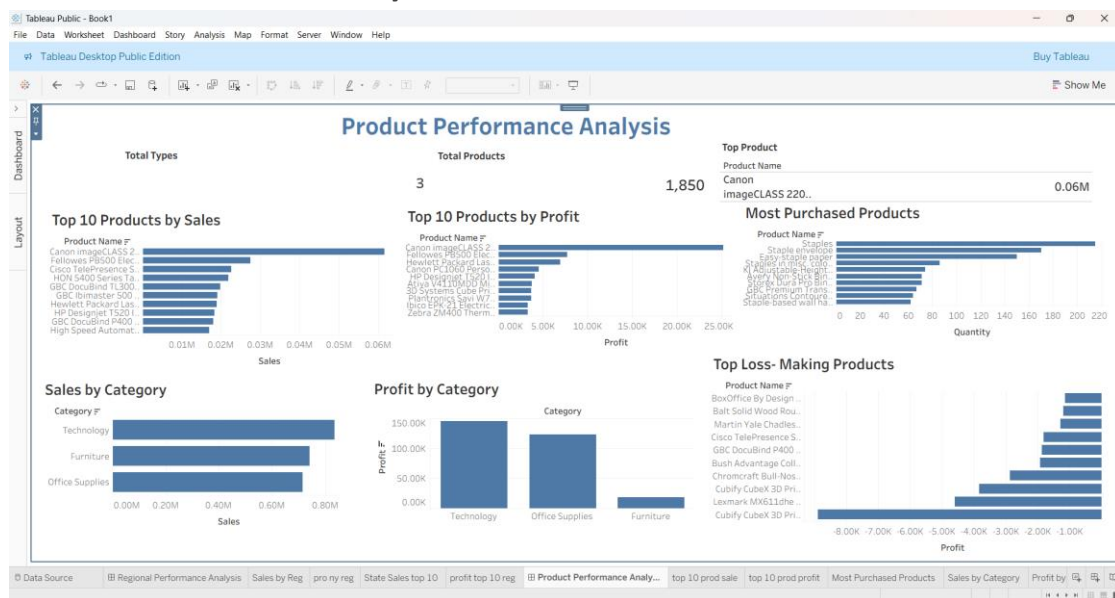
This dashboard provides detailed analysis of executive overview dashboard. It supports KPI monitoring, trend analysis and strategic decision making.

Regional Performance Analysis



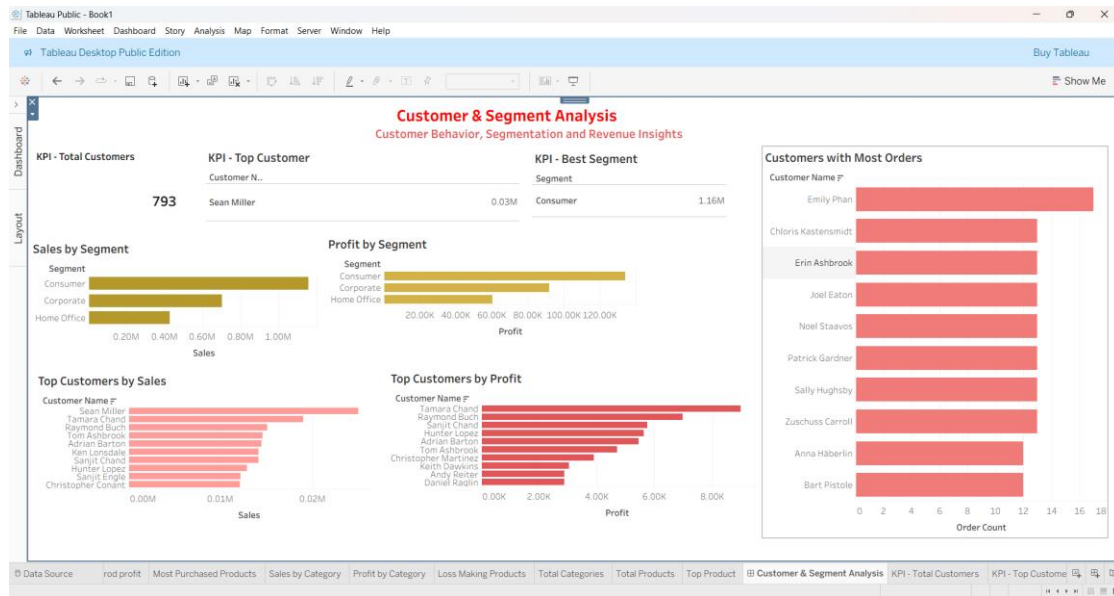
This dashboard provides detailed analysis of regional performance analysis. It supports KPI monitoring, trend analysis and strategic decision making.

Product Performance Analysis



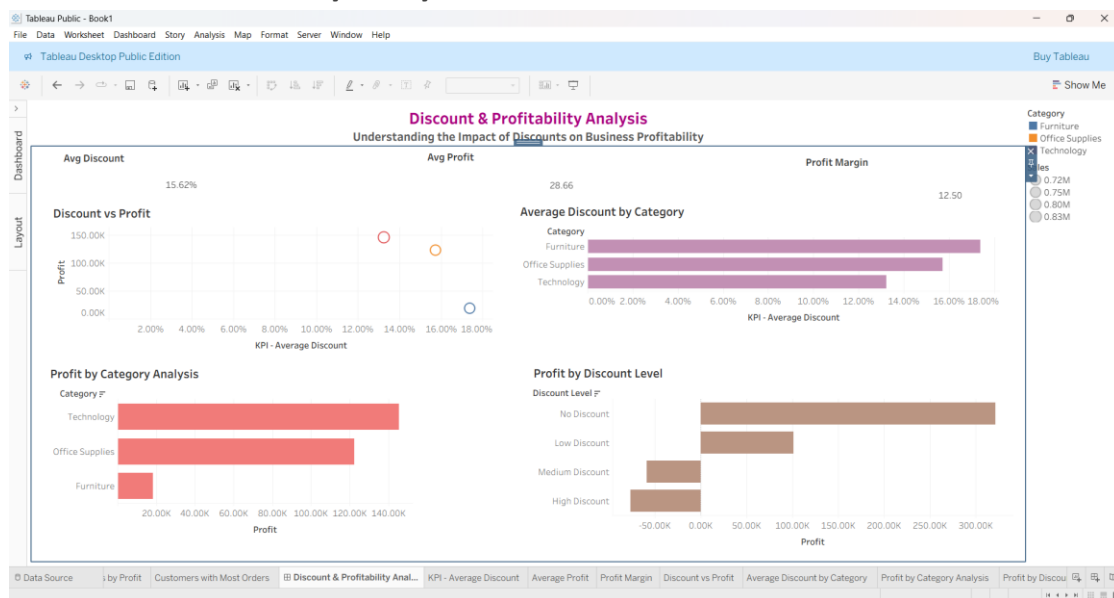
This dashboard provides detailed analysis of product performance analysis. It supports KPI monitoring, trend analysis and strategic decision making.

Customer & Segment Analysis



This dashboard provides detailed analysis of customer & segment analysis. It supports KPI monitoring, trend analysis and strategic decision making.

Discount & Profitability Analysis



This dashboard provides detailed analysis of discount & profitability analysis. It supports KPI monitoring, trend analysis and strategic decision making.